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**1. What is complete binary tree?**

**Answer:** "A complete binary tree is a binary tree with the additional property that every node must have exactly two children if an internal node, and zero children if a leaf node."

**2. How single left rotation is performed in AVL tree?**

**Answer:** This rotation is almost similar to the single right rotation. We start from tree node k1. k1 is the root while tree node k2 is its right child. Due to the left rotation, the right child k2 of the root k1 will become the root. The k1 will go down to the left child of k2.

**3. name of two divide and conquer algorithm (2 marks)**

**Answer:**

❖ Merge Sort

❖ Quick Sort

❖ Heap Sort

These three algorithms fall under _divide and conquer category.

**4. difference between call by value n call by reference?**

**Answer:**

In case of call by value, a copy of object is made and placed at the time of function calling in the activation record. By using the references, we are not making the copy but send the address of the variable that function can use the original value.

**5. describe the following**

**(i) Height of tree**

**(ii) Balance of Node**

**(i) Height of tree**

**Answer:** The height of a binary tree is the maximum level of its leaves (also called the depth).

**(ii) Balance of Node**

**Answer:** The balance of a node in a binary search tree is defined as the height of its left subtree minus height of its right subtree. In other words, at a particular node, the difference in heights of its left and right subtree gives the balance of the node**.**

**6. heap and two types of heap?**

**Answer:** Heap is a data structure of big use and benefit. It is used in priority queue. ―The definition of heap is that it is a complete binary tree that conforms to the heap order‖. Here heap order means the min and max heap. Max heap: In max heap, each node has a value greater than the value of its left and right child nodes. Moreover, the value of the root will be largest and will become lesser at downward levels. Min heap: in a (min) heap for every node X, the key in the parent is smaller than (or equal to) the key in X. This means that the value in the parent node is less than the value on its children nodes.

**7. what is binary tree brief it?**

**Answer:** "A binary tree is a finite set of elements that is either empty or is partitioned into three disjoint subsets. The first subset contains a single element called the root of the tree. The other two subsets are themselves binary trees called the left and right sub-trees".

**8. height of a tree is 5 find sum of heights (2)?**

**Answer:**

Height = 5

Nodes = 2^h = 2^5 =32

Sum of Nodes = N-(h+1)= 32-(5+1) = 32-6 = 26

**9. Here is an array with exactly 15 elements:**

**1 2 3 4 5 6 7 8 9 10 11 12 13 14 15.**

**Suppose that we are doing a binary search for an element. Indicate any elements that will be found by examining two or fewer numbers from the array.**

**Answer:** If there is an array of 15 elements, {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15} it wants to know which elements can be discovered by examining two or fewer elements. I want assume the divide and conquer aspect of a binary search, so that leads me to think these elements will be found 1, 2, 14, 15, 7, 3

rather; 1, 2, 3, 4, 6, 8, 14, 15

1 and 15 for heads and tails

2, and 14 for next elements (following or preceding)

1/2 of 15 is either 6, or 8, from 7.5

and 1/2 of 6 = 3

1/28=4

**10. What is different between binary tree and AVL tree?**

**Answer:**

An AVL tree is identical to a BST, barring the following possible differences:

❖ Height of the left and right subtrees may differ by at most 1.

❖ Height of an empty tree is defined to be (–1).

**11. Write one example of hashing?**

**Answer:** Hashing can be used to store the data. Suppose we have a list of fruits. The names of Fruits are in string. The key is the name of the fruit. We will pass it to the hash Function to get the hash key.

HashCode("apple") = 5

We pass it to the string —apple‖. Resultantly, it returns a number 5.

**12. Why queue is linear data structure?**

**Answer:** A queue is a linear data structure into which items can only be inserted at one end and removed from the other. 3 In contrast to the stack, which is a LIFO (Last In First Out) structure, a queue is a FIFO (First In First Out) structure.

**13. How do we carry out degeneration of Complete Binary Tree?**

**Answer:**

Degeneration of the (original) binary tree is mainly reflected by degeneration of the internal infix trees and less by degeneration of the external height of the PPbin tree.

**14. How we can delete a node with two Childs in a binary search tree using its right sub tree?**

**Answer:** The node to be deleted has either left child (subtree) or right child (subtree).In this case we bypass the node in such a way that we find the inorder successor of this node and then link the parent of the node to be deleted to this successor node. Thus the node was deleted.

**15. What is a skip List?**

**Answer:** A skip list for a set S of distinct (key, element) items is a series of lists S0, S1 , … , Sh such that Each list *Si* contains the special keys +∞ and -∞

• List *S*0 contains the keys of *S* in non-decreasing order

• Each list is a subsequence

of the previous one, i.e.,

$S0 \supseteq S1 \supseteq \ldots \supseteq Sh$

• List *Sh* contains only the two special keys

**16. define const keyword, reference variable, Dangling reference?**

**Answer:**

**Const keyword:**

The const keyword is used for something to be constant. The actual meanings depend on where it occurs but it generally means something is to held constant. There can be constant functions, constant variables or parameters etc.

**Reference variable:**

The references are pointers internally, actually they are constant pointers. You cannot perform many kind of arithmetic manipulation with references that you normally do with pointers.

**Dangling reference:**

The pointer of the object (when object has already been deallocated or released) is called dangling pointer.

**17. How we can generate a maze with the help of union?**

**Answer:** How can we generate maze? The algorithm is as follows:

❖ Randomly remove walls until the entrance and exit cells are in the same set.

❖ Removal of the wall is the same as doing a union operation.

❖ Do not remove a randomly chosen wall if the cells it separates are already I the same set.

**18. What is function of length () method in the Queue?**

**Answer:** In Queue length() method, has a single statement i.e. return size ;

This method returns the size of the queue, reflecting the number of elements in the queue. It is not the size of the array used internally to store the elements of the queue.

**19. Explain the two cases in which we apply double rotation in An Avl tree**

**Answer:** Sometimes a single rotation is not sufficient to balance an unbalanced tree.

The two cases are following:

1. Insertion into the right subtree of the left child of X node (RL)

2. Insertion into the left subtree of the right child of X node (LR)

**20. Write down the C++ code to implement insertion sort algorithm.**

**Answer:** Following is the code of the insertion sort in C++.

```
void insertionSort(int *arr, int N)
{
int pos, count, val;
for(count=1; count < N; count++)
{
val = arr[count];
for(pos=count-1; pos >= 0; pos--)
if (arr[pos] > val)
arr[pos+1]=arr[pos];
else break;
arr[pos+1] = val;
}
}
```

**21. How can we perform level-order traversal on a tree?**

**Answer:** We can print a binary tree level by level by employing recursive or non-recursive method.

The idea of a level-order traversal is to visit the root, then visit all nodes "1 level away" (depth 2) from the root (left to right), then all nodes "2 levels away" (depth 3) from the root, etc. For the example tree, the goal is to visit the nodes in the following order:

A level-order traversal requires using a queue (rather than a recursive algorithm, which implicitly uses a stack).

A queue is a FIFO structure, which can make the level-order traversal easier.

**22. What is Table ADT? Discuss any two implementations of table ADT. ( Marks: 5)**

**Answer:** The table, an abstract data type, is a collection of rows and columns of information.

Implementation of Table

o Unsorted Sequential Array

o Sorted Sequential Array

**23. How can the dangling reference problem be avoided?**

**Answer:** To avoid dangling reference, don't return the reference of a local variable (transient) from a function.

**24. Suppose we have the following representation for a complete Binary Search Tree, tell the Left and Right child nodes and Parent node of the node D**

**A B C D E F G H I J K L M N O P Q R S T …**

**0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 …**

**Answer:**

parent: B using formula(i/2)

left child:-H 2i=2*4=8=H

Right child:-I 2i+1=8+1=9=I

**25. How AVL is different from Binary Search Tree?**

**Answer:** A binary tree is a tree data structure in which each node has at most two children. Typically the child nodes are called left and right. One common use of binary trees is binary search trees; another is binary heaps. While an AVL tree is a self-balancing binary search tree. In an AVL tree the heights of the two child sub-trees of any node differ by at most one, therefore it is also called height-balanced. Lookup, insertion, and deletion all take O(log n) time in both the average and worst cases. Additions and deletions may require the tree to be rebalanced by one or more tree rotations.

**26. What is an Equivalent relation? Give any two examples.**

**Answer:** A binary relation R over a set S is called an *equivalence relation* if it has following properties‘:

1. Reflexivity: for all element x ȟ S, x R x

2. Symmetry: for all elements x and y, x R y if and only if y R x

3. Transitivity: for all elements x, y and z, if x R y and y R z then x R z Example; sets of peoples ,electric circuit domain.

**27. What are the properties of equivalence class?**

**Answer:** A binary relation R over a set S is called an *equivalence relation* if it has following Properties

1. Reflexivity: for all element x ∈ S, x R x

2.Symmetry: for all elements x and y, x R y if and only if y R x

3.Transitivity: for all elements x, y and z, if x R y and y R z then x R z

**28. What the maximum no of comparisons we have to perform during insertion in BST?**

**Answer:** If we have a complete binary tree with n numbers of nodes, the depth d of the tree can be found by the following equation: d=log2 (n + 1) – 1. If the tree is complete binary or near-to-complete, we need log2(n) number of comparison. Whereas in a linked list, the comparisons required could be a maximum of n**.**

**29. How heap sort works to set a set of data?**

**Answer:** Min heap and max heap are used to sort the data. Min heap convert the data in ascending sorted order and max heap convert the in descending order.

**30. How can we search an element in skip list? 2Marks**

**Answer:** We search for a key x in the following fashion:

We start at the first position of the top list

•At the current position p, we compare x with y key(after(p))

• x = y: we return element(after(p))

• x > y: we —scan forward‖

• x < y: we —drop down‖

•If we try to drop down past the bottom list, we return NO_SUCH_KEY

**31. What is Function Call Stack Give short answer?**

**Answer:** Stacks play a key role in implementation of function calls in programming languages. In C++, for example, the "call stack" is used to pass function arguments and receive return values. The call stack is also used for "local variables"

**32. What is an ADT?**

**Answer:** The table, an abstract data type, is a collection of rows and columns of information.

**33. What are the applications of Binary Tree?**

**Answer:** The binary tree is a useful data structure for rapidly storing sorted data and rapidly retrieving stored data. It can hold objects that are sorted by their keys. It can represent a structured object, an operating system maintains a disk's file system as a tree, where file folders act as tree nodes.

**34. How we can implement Table ADT using Linked?**

**Answer:** List linked list is one choice to implement table abstract data type. For unsorted elements, the insertion at *front* operation will take constant time. But if the data inside the list is kept in sorted order then to insert a new element in the list, the entire linked list is traversed through to find the appropriate position for the element.

**35. Give the names of basic Queue Operations**

**Answer:**

**Queue Operations**

The queue data structure supports the following operations:

**Operation Description**

enqueue(X) Place X at the *rear* of the queue.

dequeue() Remove the *front* element and return it.

front() Return *front* element without removing it.

IsEmpty () Return TRUE if queue is empty, FALSE otherwise

**36. If we allow assignment to constants what will happen?**

**Answer:** Suppose _*a*' has the value number 3. Now we assigned number 2 the number 3 i.e. all the number 2 will become number 3 and the result of *2 + 2* will become 6. Therefore it is not allowed.

**37. Explain the process of Deletion in a Min-Heap**

**Answer:** We have to write a function deleteMin(), which will find and delete the minimum number from the tree. Finding the minimum is easy; it is at the top of the heap.

**38. Give one benefit of using Stack.**

**Answer:** In computer science, a stack is a last in, first out (LIFO) abstract data type and data structure. A stack can have any abstract data type as an element, but is characterized by only two fundamental operations: push and pop. the data structure itself enforces the proper order of calls.

**39. Why List wastes less memory as compared to Arrays.**

**Answer:**

1. Linked lists do not need contiguous blocks of memory; extremely large data sets stored in an array might not be able to fit in memory.

2. Linked list storage does not need to be pre_allocated (again, due to arrays needing contiguous memory blocks).

3. Inserting or removing an element into a linked list requires one data update, inserting or removing an element into an array requires n (all elements after the modified index need to be shifted). Array is a collection of same data type. In linked list there are two field one is address and other is pointer. In array elements are arranged in a specific order.

**40. Why we can change the size of list after its creation when we can not do that in simple arrays?**

**Answer:** Somehow the answer will be same as part 1 because Inserting or removing an element into a linked list requires one data update, inserting or removing an element into an array requires n (all elements after the modified index need to be shifted).Array is a collection of same data type. The size of array is mentioned with its declaration. in arrays the elements are in contiguous position. One must after another. While in linked list we gave the address of next element in the next part of node.

**41. Give any three characteristics of Union by Weight method.**

**Answer:** Following are the salient characteristics of this method:

❖ Maintain sizes (number of nodes) of all trees, and during *union*.

❖ Make smaller tree, the sub-tree of the larger one.

❖ Implementation: for each root node i, instead of setting parent[i] to -1, set it to -k if tree rooted at i has k nodes.

**42. "For smaller lists, linear insertion sort performs well, but for larger lists, quick sort is suitable to apply." Justify why?**

**Answer:** Since for smaller lists number of comparisons and interchange (if needed) decreases radically in insertion sort than quick sort. For larger lists, opposite occurs in both sorting techniques.

**43. Describe any two uses of priority queues?**

**Answer:** Priority queue being used at many places especially in the operating systems. In operating systems, we have queue of different processes. If some process comes with higher priority, it will be processed first. Here we have seen a variation of queue. We will use the priority queue in the simulation. The events will be inserted in the queue and the event going to occur first in future, will be popped.

**44. What is Disjoint Sets? Explain with an example.**

**Answer:**

**Disjoint sets**

Two sets are disjoint if they have no elements in common. Example:-Suppose there are many people around you. How can we separate those who are related to each other in this gathering? We make groups of people who are related to each other. The people in one group will say that they are related to each other. Similarly we can have many groups. So every group will say that they are related to each other with family relationship and there is no group that is related to any other group. The people in the group will say that there is not a single person in the other group who is related to them.

**45. Describe the Use of Queues?**

**Answer:**

**Use of Queues:**

Out of the numerous uses of the queues, one of the most useful is *simulation*. A simulation program attempts to model a real-world phenomenon. Many popular video games are simulations, e.g., SimCity, Flight Simulator etc. Each object and action in the simulation has a counterpart in the real world. Computer simulation is very powerful tool and it is used in different high tech industries, especially in engineering projects. For example, it is used in aero plane manufacturing. Actually Computer Simulation is full-fledged subject of Computer Science and contains very complex Mathematics, sometimes. For example, simulation of computer networks, traffic networks etc.

**46. How we evaluate postfix expressions?**

**Answer:**

**Evaluating Postfix Expression:**

o Each operator in a postfix expression refers to the previous two operands.

o Each time we read an operand, we push it on a stack.

o When we reach an operator, we pop the two operands from the top of the stack, apply the operator and push the result back on the stack.

**47. Give the difference between strict and complete binary tree.**

**Answer:** A binary tree said to be a strictly binary tree if every non-leaf node in a binary has non-empty left and right sub trees. If there are left and right sub trees for each node in the binary tree as shown a complete binary tree.

**48. What is meant by an empty stack?**

**Answer:** Empty stack means that there is no element on the stack. IsEmpty() boolean function returns true if stack is empty, false otherwise.

**49. A complete binary tree can be stored in an array. While storing the tree in an array we leave the first position (0th index) of the array empty. Why?**

**Answer:** In case of having a node with left and right children, stored at position i in the array, the left child will be at position 2i and the right child will be at 2i+1 position. If the value of i is 2, the parent will be at position 2 and the left child will be at position 2i i.e. 4 .The right child will be at position 2i+1 i.e. 5. we have not started the 0 th position. It is simply due to the fact if the position is 0, 2i will also become 0. So we will start from the 1 st position, ignoring the 0 th .

**50. Give the effect of sorted data on Binary Search.**

**Answer:** The binary tree is a fundamental data structure used in computer science. The binary tree is a useful data structure for rapidly storing sorted data and rapidly retrieving stored data.

**51. Give your comment on the statement that heap uses least memory in array representation of binary trees. Justify your answer in either case.**

**Answer:** A heap is a complete binary tree, so it is easy to be implemented using an array representation It is a binary tree with the following properties:-

❖ Property 1: it is a complete binary tree

❖ Property 2: the value stored at a node is greater or equal to the values stored at the children

❖ From property 2 the largest value of the heap is always stored at the root.

**52. Is the following statement correct? If your answer is No, then correct it.**

**"A tree is an AVL tree if at least half of the nodes fulfill the AVL condition"**

**Answer:** Statement is wrong, Correct statement is:

A tree is an AVL tree if all the nodes fulfill the AVL condition.

**53. Explain the following terms:**

**1. Collision**

**2. Linear Probing**

**3. Quadratic Probing**

**Answer:**

**Collision:**

When two values hash to the same array location, this is called a *collision*..

**Linear Probing**

When there is a collision, some other location in the array is found. This is known as linear probing. In linear probing, at the time of collisions, we add one to the index and check that location. If it is also not empty, we add 2 and check that position. Suppose we keep on incrementing the array index and reach at the end of the table. We were unable to find the space and reached the last location of the array.

**Quadratic Probing**

In the quadratic probing when a collision happens we try to find the empty location at index + 1^2. If it is filled then we add 2^2 and so on. Quadratic probing uses different formula: 1. Use F(i) = i 2 (square of i) to resolve collisions

2. If hash function resolves to H and a search in cell H is

inconclusive, try H+12 ,H+22,H+32

**54. How we can delete a node with two Childs in a binary search tree using its right sub tree.**

**Answer:**

The node to be deleted has either left child (sub-tree) or right child (sub-tree).In this case we bypass the node in such a way that we find the in-order successor of this node and then link the parent of the node to be deleted to this successor node. Thus the node was deleted.

**55. Why we use Reference Variables. Give one example.**

**Answer:**

C++ references allow you to create a second name for the a variable that you can use to read or modify the original data stored in that variable.

```
#include
main()
{
int x;
int& foo = x;
// foo is now a reference to x so this sets x to 56
foo = 56;
std::cout << x <<std::endl;
}
```

**56. What are different applications of Hashing?**

**Answer :**

❖ Compilers use hash tables to keep track of declared variables (symbol table).

❖ A hash table can be used for on-line spelling checkers — if misspelling detection (rather than correction) is important, an entire dictionary can be hashed and words checked in constant time.

❖ Game playing programs use hash tables to store seen positions, thereby saving computation time if the position is encountered again.

**57. Give the operation names that we can perform on Table abstract data type.**

**Answer:**

Insert

Find

Remove.

**58. Define Reference Variable, Dangling Reference & Const?**

**Answer:** In the C++ programming language, a reference is a simple reference datatype that is less powerful but safer than the pointer type inherited from C. The name C++ reference may cause confusion, as in computer science a reference is a general concept datatype, with pointers and C++ references being specific reference datatype implementations Dangling Reference & Const Dangling pointers and wild pointers in computer encoding are pointers that do not point to a valid object of the suitable type. These are special cases of violations of memory safety.

**59. What is the use of Reference Variable, Give example?**

**Answer:** A reference variable is used to refer a particular object Location which resides on a garbage collectable heap. C++ supports one more type of variable called reference variable. In addition to pointer variable and value variable of reference variable behaves similar to both value variable and pointer variable. Reference variable has all access that variable has.

**60. Give your comments on the statement:**

**"Efficiently developed data structures decrease programming effort"**

**Answer:**

- Reduces programming effort by providing useful data structures and algorithms so you don't have to write them yourself.
- Increases performance by providing high-performance implementations of useful data structures and algorithms. Because the various implementations of each interface are interchangeable, programs can be easily tuned by switching implementations.

**61. Which implementation of disjoint set is better with respect space and why?**

**Answer:** An array that is a single dimensional structure and seems better with respect to space. Suppose we have 1000 set members i.e. names of people. If we make a Boolean matrix for 1000 items, its size will be 1000 x 1000. Thus we need 1000000 locations for Boolean items, its size will be 1000 x 1000. Thus we need 1000000 locations for Boolean values. in case of an array the number of location will be 1000. Thus this use of array i.e. tree like structure is better then two dimensional arrays in term of space to keep disjoint sets and doing union and find operation.

**62. What is Table ADT? Discuss any two implementations of table ADT.**

**Answer:** The table, an abstract data type, is a collection of rows and columns of information.

Implementation of Table

- Unsorted Sequential Array
- Sorted Sequential Array

**63. If a Binary Tree has N internal nodes what are the no. of external nodes in it.**

**Answer:** The No. of external nodes will be N+1.

**64. Which is Dynamic Equivalence Problem with an example?**

**Answer:** We are using sets to store the elements. For this purpose, it is essential to remember which element belongs to which set. We know that the elements in a set are unique and a tree is used to represent a set. Each element in a tree has the same root, so the root can be used to name the set. The item (number) in the root will be unique as the set has unique values of items. We use this root as the name of set for our convenience. Otherwise, we can use any name of our choice. So the element in the root (the number) is used as the name of the set. The *find* operation will return this name. Due to the presence of many sets, there will be a collection of trees. In this collection, each tree will be representing one set. If there are ten elements, we will have ten sets initially. Thus there will be ten trees at the beginning. In general, we have N elements and initially there will be N trees. Each tree will have one element. Thus there will be N trees of one node. Now here comes a definition for this collection of trees, which states that a collection of trees is called a *forest*. The trees used for the sets are not necessarily binary trees. That means it is necessary that each node should have a maximum of two children nodes. Here a n may have more than two children nodes.To execute the union operation in two sets, we merge the two trees of these sets in such a manner that the root of one tree points to the root of other. So there will be one root, resulting in the merger of the trees. In the find operation, when we call *find (x)*, it helps us to know which set this *x* belongs to. Internally, we find this *x* in a tree in the *forest*. When this *x* is found in a tree the *find* returns the number at root node (the name of the set) of that tree.

**65. "is a sibling of" set of all human beings is equivalence relation or not? Explain.**

**Answer:** Yes, it is an equivalence relation. First rule is reflexive i.e. for all element *x S, x R x*. Suppose that *x* is Haris so *Haris R Haris*. This is true because everyone is related to each other. Second is Symmetry: for all elements *x* and *y, x R y* if and only if *y R x*. Suppose that *y* is Saad. According to the rule, *Haris R Saad* if and only if *Saad R Haris*. If two persons are related, the relationship is symmetric i.e. if I am cousin of someone so is he. Therefore if Haris is brother of Saad, then Saad is certainly the brother of Haris. The family relationship is symmetric. This is not the symmetric in terms of respect but in terms of relationship. The transitivity is: ‗for all elements *x*, *y* and *z*. If *x R y* and *y R z*, then *x R z*'. Suppose *x* is Haris, *y* is Saad and *z* is Ahmed. If Haris ―is related to‖ Saad, Saad ―is related to‖ Ahmed. We can deduce that Haris ―is related to‖ Ahmed. This is also true in relationships. If you are cousin of someone, the cousin of that person is also related to you. He may not be your first cousin but is related to you.

**66. Linked Memory?**

Linked memory is a memory in which the various cells of memory are not located continuously. In this process, each cell of the memory not only contains the value of the element but also the information where the next element of the list is residing in the memory. It is not necessary that the next element is at the next location in the memory. It may be anywhere in the memory. We have to keep a track of it. Thus, in this way, the first element must explicitly have the information about the location of the second element. Similarly, the second element must know where the third element is located and the third should know the position of the fourth element and so on.

**67. Code for removing an element from a linked list data structure if it is implementation by using linked memory methods?**

**Answer:**

```
void remove() {
if( currentNode != NULL && currentNode !=
headNode) { lastCurrentNode-
>setNext(currentNode->getNext()); delete
currentNode;
currentNode = lastCurrentNode->getNext();
```

```
size--;
}
}
```

**Best of Luck**

*For Free help, more Mid/Final term Files and for Exam Preparation*

**:You Can Join:**